THE QUOTIENT-STACK SQUARE OF A GROUPOID MORPHISM IS A 2-FIBRE PRODUCT

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-groupoids.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original lemma 046Q: sources/stacks-chow/tags.txt, source lines 5481–5481. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 044R: sources/stacks-chow/tags.txt, source lines 5410–5410. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04Y2: sources/stacks-chow/tags.txt, source lines 6415–6415. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Notation and assumption as in Lemma 046Q. Let $(U'', R'', s'', t'', c'')$ be the groupoid in algebraic spaces over B constructed above. There is a 2-commutative square*

$$\begin{array}{ccc} [U''/R''] & \xrightarrow[[g]]{} & [U/R] \\ \downarrow & & \downarrow{\scriptstyle [f]} \\ \mathcal{S}_{U'} & \longrightarrow & [U'/R'] \end{array}$$

which identifies $[U''/R'']$ with the 2-fibre product.

Proof. The maps $[f]$ and $[g]$ come from an application of Lemma 046Q and the other two maps come from Lemma 044R (and the fact that $(U'', R'', s'', t'', c'')$ lives over U'). To show the 2-fibre product property, it suffices to prove the lemma for the diagram

$$\begin{array}{ccc} [U''/_pR''] & \xrightarrow[[g]]{} & [U/_pR] \\ \downarrow & & \downarrow{\scriptstyle [f]} \\ \mathcal{S}_{U'} & \longrightarrow & [U'/_pR'] \end{array}$$

of categories fibred in groupoids, see Stacks, Lemma 04Y2. In other words, it suffices to show that an object of the 2-fibre product $\mathcal{S}_U \times_{[U'/_pR']} [U/_pR]$ over T corresponds to a T-valued point of U'' and similarly for morphisms. And of course this is exactly how we constructed U'' and R'' in the first place.

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In detail, an object of $\mathcal{S}_U \times_{[U'/_pR']} [U/_pR]$ over T is a triple (u', u, r') where u' is a T-valued point of U', u is a T-valued point of U, and r' is a morphism from u' to $f(u)$ in $[U'/R']_T$, i.e., r' is a T-valued point of R with $s'(r') = u'$ and $t'(r') = f(u)$. Clearly we can forget about u' without losing information and we see that these objects are in one-to-one correspondence with T-valued points of R''.

Similarly for morphisms: Let (u'_1, u_1, r'_1) and (u'_2, u_2, r'_2) be two objects of the fibre product over T. Then a morphism from (u'_2, u_2, r'_2) to (u'_1, u_1, r'_1) is given by $(1, r)$ where $1 : u'_1 \to u'_2$ means simply $u'_1 = u'_2$ (this is so because $\mathcal{S}_U$ is fibred in sets), and r is a T-valued point of R with $s(r) = u_2$, $t(r) = u_1$ and moreover $c'(f(r), r'_2) = r'_1$. Hence the arrow

$$(1, r) : (u'_2, u_2, r'_2) \to (u'_1, u_1, r'_1)$$

is completely determined by knowing the pair (r, r'_2). Thus the functor of arrows is represented by R'', and moreover the morphisms s'', t'', and c'' clearly correspond to source, target and composition in the 2-fibre product $\mathcal{S}_U \times_{[U'/_pR']} [U/_pR]$. $\square$